

Decision Making Under Uncertainty: 95-760

Class project

Due: Tuesday, October 6th, 2026

The project for this class is an opportunity to explore optimization concepts in a topic of your interest. You will work in teams of up to **three** students to either develop an optimization model for a real-world problem of your choice or explore the use of Generative AI (GenAI) in optimization.

WHAT IS THE PROJECT ABOUT?

You have two options for your project. You will work in teams and select **one** of the following options.

OPTION 1: OPTIMIZATION IN A TOPIC OF YOUR INTEREST

Identify a real-world problem that interests your team and formulate it as an optimization problem. The problem can come from any field or application area. The goal of this option is to develop an optimization model that could potentially be used to support a real-world decision. You do not need to solve the problem if appropriate data are not available. Instead, focus on clearly defining the problem and developing an optimization model.

Your project should identify the decision variables, parameters, objective function, and constraints of the problem. You should also discuss the assumptions and limitations of your model and explain how the model could be used to support decision-making. If you don't have the data, you could simply define your parameters to write your constraints. For example, if you don't have the cost, you could say c_i is the cost for resource i .

As part of the project, you must provide a **concrete example of a real-world situation in which your model could be used**. Explain what motivated your project and what decision-maker or organization could benefit from using your model. Your problem should have at least five different constraints and you should discuss two different objective functions.

GenAI may not be used for Option 1. The optimization problem, formulation, parameters, and analysis must be developed by your team.

OPTION 2: GENAI AND OPTIMIZATION

Explore how Generative AI can be used to develop and solve optimization problems. In this option, you will interact with a GenAI tool and critically evaluate its responses.

For example, you could ask GenAI to act as an optimization expert and ask you questions about your optimization formulation. You could also ask GenAI to develop parameters for a problem and solve the resulting model.

Your project should highlight what you learned from these interactions. In particular, discuss:

- What did GenAI do well?
- Where did GenAI make mistakes or provide misleading answers?
- What aspects of optimization did GenAI explain particularly well?
- Where was human judgment or expertise necessary?
- What did you learn about optimization from working with GenAI?

You should not simply present the answers generated by GenAI. The goal of this project is to **critically evaluate** how GenAI can and cannot support the optimization modeling process.

You must also provide a **concrete example of a real-world situation where the optimization model developed or analyzed in your project could be used**. Explain what motivated your project and what decision-maker or organization could benefit from using your model.

NOTE: The goal of the project is not to develop the largest or most complicated optimization model possible. A small, well-defined model that clearly represents an interesting real-world decision is preferable to a large model that is difficult to explain or justify.

DELIVERABLES

The project has three required deliverables: a project proposal, an in-class presentation, and a recording of your presentation. There is also an optional opportunity to receive feedback during the project. See details of each assignment below:

DELIVERABLE 1: TEAMS – DUE TUESDAY, SEPTEMBER 22ND AT 11:59 PM

Submit a PDF to Gradescope with the following information:

- Team members
- Project option you are doing: Option 1 or Option 2
- A brief description of your proposed topic/project
- Any questions you have about the project

Your topic does not need to be finalized at this point. The purpose of this submission is to make sure that you have identified a team and a direction for your project. You cannot change your team after this submission.

OPTIONAL FEEDBACK ASSIGNMENT – DUE FRIDAY, SEPTEMBER 25TH AT 11:59 PM

If you would like feedback throughout your project, you may submit a PDF to Gradescope answering the following questions:

- a Which project option did you select?
- b What is the problem or question you are planning to explore?
- c What are the decisions that need to be made in your problem?
- d What do you think your objective(s) will be?
- e What are some of the constraints or limitations of the problem?
- f What questions or concerns do you have about your project?

Note that you can answer as many or as few of the above questions. I will provide feedback on your proposed approach. This assignment is optional and will not affect your project grade. I am also happy to meet with you to discuss your submission.

IN-CLASS PRESENTATION – TUESDAY, OCTOBER 6TH AT 9:30 AM (SECTION A) OR 2:00 PM (SECTION B)

You will present your project to your classmates. To encourage discussion and allow everyone to learn from the different projects, presentations will take place in small groups.

Each team will present their project twice to some of their classmates. The length of each presentation will depend on the number of teams. This information will be posted on Canvas by Thursday, September 24th.

You should be prepared to clearly explain your problem, your optimization model and/or your main findings, and the insights you gained from the project. The format of the presentation is up to your team. However, we recommend using slides to help organize your presentation and communicate your ideas clearly.

RECORDING OF YOUR PRESENTATION – TUESDAY, OCTOBER 6TH AT 11:59 PM

Because the TAs and I (Prof. Barrios) will not be able to attend all of the in-class presentations, each team must submit a recording of their presentation.

The recording should not exceed 10 minutes. **All team members must participate and present part of the project. Each team member's camera must be on while they are presenting.** The recording should cover the same material presented during the in-class presentation. If there were any questions or comments during the in-class presentation, you may address them or incorporate them into your video.

HOW WILL YOU BE GRADED?

The project grade will be divided into three components:

- **In-class presentation (10%):** Your grade for the in-class presentation will be determined through peer review.
- **Peer review (10%):** Each student (not each team) will complete a peer evaluation for the projects they observe during class. The peer review should be submitted by the end of class. Your grade will be based on the quality and thoughtfulness of your feedback.
- **Video recording (80%):** The video recording will be graded using the rubric provided below.

Problem and Motivation	Clearly explains a meaningful real-world problem, what motivated the project, and how optimization can support the decision.
Optimization Model / Application	Correctly and clearly applies optimization concepts. The model, decisions, objective, and constraints are appropriate for the problem.
Analysis and Insights	Goes beyond presenting the model/results and provides thoughtful analysis and meaningful insights.
Assumptions and Limitations	Clearly identifies important assumptions and limitations and discusses how they affect the model, results or, if option 2 is selected, what are the limitations of using GenAI.
Communication and Organization	Overall quality of the presentation. Presentation is clear, well organized, engaging, and easy to follow. Visuals effectively support the presentation. The presentation is not text heavy nor has multiple screenshots from Excel/GenAI.

OPTION-SPECIFIC EXPECTATIONS:

- **Option 1:** The optimization model should appropriately represent the real-world problem. The formulation should clearly define the decision variables, parameters, objective function, and constraints.
- **Option 2:** The project should critically evaluate the use of GenAI in optimization. Simply presenting GenAI-generated responses is not sufficient. The project should provide concrete examples of both the strengths and limitations of GenAI and discuss where human judgment or expertise is necessary. Make sure to mention with GenAI model(s) you used.